

# Mahdi Heydari

## Protocol Engineer & Security Researcher

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Protocol engineer with 15+ years in software and 8+ in decentralized systems. I design consensus, threshold-signature and custody protocols, prototype them, then ship them on audited production cores. Security researcher focused on consensus and networking code.

### HIGHLIGHTS

- **\$30,000 Aptos bounty** for a High-severity consensus bug that could halt the network.
- **Designed four decentralized systems:** the BrightID node, the Muon threshold oracle, the Zellular sequencer and Zex's multi-chain FROST custody.
- **1st place, EigenLayer AVS MicroHacks 2024** (\$12K, 100+ builders) and a 100,000 EIGEN ecosystem grant.
- **About 1M orders/s** from a C matching engine prototype, up from a 100K target, verified against Python by state digests.

### EXPERIENCE

**Zex · Founder & Protocol Architect → Tech Lead & Reviewer** 2024 – present

Multi-chain exchange run as a replicated state machine. Designed FROST custody across Bitcoin, EVM, Tron, Solana and Aptos with an independent co-signing key, wallet-free login, and the move to Firedancer's Alpenglow consensus.

**Zellular · Founder & Protocol Architect** 2023 – 2025

Consensus and custody as services for dApps. Designed a leader-based BFT sequencer run as an EigenLayer AVS and wrote the pyfrost threshold-signature spec. Lead author of the Python EigenLayer SDK.

**Muon Network · Architect → Technical Lead** 2021 – 2023

Designed a threshold-signature oracle network (1,500+ node mainnet) and led a lean Python FROST node to replace a complex JavaScript one.

**BrightID · Core Team Developer** 2018 – 2023

Primary architect of the node for a 73K+ user identity network: consensus through a dedicated PoA chain, anti-sybil research and blind-signature verifications.

**Freelance · Two-person team** 2013 – 2018

220+ projects at 4.8 / 5 in Python, data and, from 2017, Ethereum.

**City transit systems, Shiraz · Software Developer** 2010 – 2013

Bus fare payments for hundreds of thousands of riders, with two independent settlement paths that had to agree.

### SECURITY RESEARCH

Aptos (\$30K, plus 3 valid duplicates) · Sui (8 DoS reports, 4 reachable by unauthenticated peers) · Zex internal audit (17 findings across FROST signing and multi-chain deposits, several critical) · Muon (rogue-key and DoS flaws in the DKG).

### SKILLS

BFT consensus, sequencing, FROST, DKG, Schnorr, BLS, EigenLayer AVS · Consensus and network security review · Python, TypeScript, Solidity · AI-assisted development and auditing

### EDUCATION

B.S. Software Engineering, Shiraz University, 2010